Practical - 5

Mayur Zope TE Comp A 71

Write a program to demonstrate Sub-netting and find subnet masks

```
#include <iostream>
using namespace std;

int main() {
    int cidr;
    int subnet[4] = {0, 0, 0, 0}; // Subnet mask in 4 octets

    cout << "Enter CIDR value (e.g., 24): /";
    cin >> cidr;

    if (cidr < 0 || cidr > 32) {
        cout << "Invalid CIDR. Must be between 0 and 32." << endl;
        return 1;
    }

    // Fill subnet mask with 1s
    for (int i = 0; i < cidr; i++) {
        subnet[i / 8] += (1 << (7 - (i % 8)));
    }

    cout << " 📡 Subnet Mask: ";
    for (int i = 0; i < 4; i++) {
        cout << subnet[i];
        if (i < 3) cout << ".";
    }
    cout << endl;

    return 0;
}
```

// OUTPUT

```
lab314@lab314-ThinkCentre-M70s: ~/Documents/nmiet
base) lab314@lab314-ThinkCentre-M70s:~/Documents/nmiet$ g++ CNS.cpp
base) lab314@lab314-ThinkCentre-M70s:~/Documents/nmiet$ ./a.out
nter CIDR value (e.g., 24): /20
 Subnet Mask: 255.255.240.0
base) lab314@lab314-ThinkCentre-M70s:~/Documents/nmiet$
```